MILITARY SPECIFICATION

LINK, CARTRIDGE, 20MM: XM24

1. SCOPE

1.1 This specification covers an articulated, metallic link for assembling 20MM cartridges into flexible, disintegrating belts for feeding the 20MM Automatic Gun M139.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116 - Preservation, Methods of
MIL-W-13855 - Weapons, Small Arms, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-C-45662 - Calibration System Requirements

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition

DRAWINGS

U. S. Army Weapons Command

A7309999 - General Data Governing Protective Finish
B8769067 - Magnetic Particle Inspection
C11577416 - Link, Cartridge, 20MM: XM24

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

U. S. Army Weapons Command

IEL 11577416 - Index of Inspection Equipment Lists for Link, Cartridge, 20MM: XM24

PDS 11577416 - Packaging Data Sheet for Link, Metallic Belt, 20MM: XM24

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM E18 - Methods of Rockwell Hardness and Rockwell Superficial Hardness

(Applications for copies should be addressed to: The American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Quality evaluation samples. On each contract, samples of links representing initial product shall be forwarded to a testing agency designated by the procuring activity (see 6.1.5). Samples shall be representative of material which has been inspected and determined to be acceptable. One of the link packs shall be considered as the pilot pack.

3.2 Material and construction. The links shall conform to this specification, to drawing C11577416, and to the applicable provisions of MIL-W-13855.

3.2.1 Twisting. A belt assembly of 12 links filled with dummy or inert rounds of ammunition shall be held vertically and shall be capable of being twisted in both directions until the last cartridge forms an angle of 180° with the first cartridge. The angles between individual cartridges shall be approximately equal.

3.2.2 Fanning. The same filled belt assembly as 3.2.1 placed on a flat surface shall be capable of being fanned manually without difficulty in either direction so that the noses, when pointed inward, form a semicircle having a maximum radius of 7.1 inches and the bases, when pointed inward, form a semicircle having a maximum radius of 11.8 inches.

3.2.3 Elongation. The same filled belt assembly as 3.2.1 shall be capable of withstanding a pull of 150 ± 5 pounds for two minutes without fracture,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

permanent deformation or a permanent set exceeding 0.125 inch. The belt shall be subjected to an initial tension load of 20 ± 2 pounds before and after each pull test.

3.2.4 Foldover. Links filled with dummy or inert rounds of ammunition and assembled into belts shall fold over freely. In addition, there shall be no interference between the projecting surfaces of the links.

3.2.5 Hinging. With the links filled with dummy or inert rounds and assembled into belts, the single loop shall move freely in the double loop of the preceding link and shall turn readily on the cartridge held by the double loop.

3.2.6 Functioning. To assure proper functioning links shall be capable of being assembled with live 20MM practice cartridges into belts, fired, then cleaned, reassembled and refired three times without malfunctioning, cracking, breaking or separating.

3.3 Marking. The marking required by the drawings shall be applied by stamping or engraving before application of the protective finish.

3.4 Protective finish. The protective finish shall be evenly deposited, free from bare spots, pits or scratches exposing the metal and shall not have a mottled appearance.

3.5 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855 and the following requirements:

3.5.1 Defects in metal. The links shall not show laminations, inclusions, voids, cuts or tears extending through the thickness of the metal or having any dimension exceeding the metal thickness.

3.5.2 Defects of form. The links shall be completely formed and free from dents and other deformations detrimental to functioning.

3.5.3 Sharp edges and burrs. The links shall be free from burrs, sharp edges or corners, and any other condition that will scratch or remove metal from the cartridge, the feed mechanism, the flexible and fixed feed chutes or the ejector chute or cause injury to personnel in handling.

3.5.4 Cleanliness. Completed links shall be free from corrosion, chips or any foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality evaluation sample inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.4 Quality evaluation sample inspection. Inspection of the quality evaluation samples (see 3.1) and the pilot packs (see 5.1) shall be performed at the designated testing agency. Quality evaluation samples and pilot packs shall be subject to the inspection specified herein and such other inspection as is necessary for compliance with the requirements of the contract. Failure of the sample to comply with the requirements of the drawings and the specification shall result in sample disapproval.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. The formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions, and contractor's delivery schedule. Each lot shall contain links from not more than one heat of steel unless physical characteristics have been previously checked on a heat-treat lot or batch basis, in which case the lot may contain links made from more than one heat of steel.

4.5.1.1 Lot identification. Each link and each packed lot shall be identified in accordance with the applicable drawings and MIL-STD-1168.

4.5.2 Examination. Examination of links and packaging shall be performed in accordance with the examination provisions of MIL-W-13855.

4.5.2.1 Links. Sample size shall be in accordance with MIL-STD-105.

4.5.2.2 Packaging. Sample size shall be in accordance with MIL-STD-105, using inspection level I.

4.5.3 Classification of defects. Classification of defects and acceptable quality levels (AQL's) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

a. The AQL listed for each defect shall be applied to the individual defect, not to a group of defects.

b. Listings under "Zone" are given only as an aid in locating applicable dimensions or values.

c. The presence of nicks, burrs, scratches, tool marks, deformations, missing operations, or other signs of poor workmanship should be considered. Samples should be compared with a specimen of known acceptable quality.

d. Where "visual" is specified as the method of dimensional inspection, the samples shall be either scaled or compared with a specimen of known acceptable quality.

e. Where SME (Standard Measuring Equipment) is specified as the method of inspection, the characteristic shall be inspected with commercial or standard inspection equipment, subject to the approval of the Government representative.

f. Examination of packaging defects specified in 4.5.3.4 shall apply to each link or container as applicable.

4.5.3.1 Link, Cartridge: Drawing C11577416

Categories	Defects	AQL	Method of inspection
CRITICAL			
None defined			
Major			
101	Center distance	0.65	8778537
102	Diameter, double loop (2)	0.65	SME
103	Diameter, single loop	0.65	SME
Minor			
201	Width, tab clearance	1.5	8778471
202	Width, slot	1.5	8778470
203	Width, overall	1.5	8778469
204	Marking; missing, incorrect or illegible	1.5	Visual
205	Protective finish; missing or defective	1.5	Visual
206	Workmanship (except as otherwise classified)	1.5	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

4.5.3.2 Loop, Double: Drawing D11577414

Categories	Defects	AQL	Method of inspection
CRITICAL			
None defined			
Major			
101	Diameter, inside (2)	0.65	SME
102	Thickness	0.65	SME
103	Diameter, ball; true position requirement	0.65	SME
104	Height -A-	0.65	SME
105	Width -B-	0.65	SME
106	Location, ball	0.65	SME
107	Indentation and true position requirements	0.65	SME
108	Location, relief	0.65	SME
Minor			
201	Marking	1.5	Visual
202	Requirements and location, ends	1.5	SME
203	Requirements, intersection	1.5	SME
204	Radius, ball	1.5	SME
205	Workmanship (except as otherwise classified)	1.5	Visual

4.5.3.3 Loop, Single: Drawing D11577415

Categories	Defects	AQL	Method of inspection
Major			
101	Profile link enclosure	0.65	SME
102	Diameter, inside	0.65	SME
103	Thickness	0.65	SME
104	Location, ball	0.65	SME
105	Location, ends	0.65	SME
106	Requirements, cut out	0.65	SME

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

Categories	Defects	AQL	Method of inspection
Minor			
201	Marking	1.5	Visual
202	Overall width	1.5	SME
203	Flange profile, functional	1.5	8778477
204	Workmanship (except as otherwise classified)	1.5	Visual

4.5.3.4 Packaging. Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet PDS 11577416 (Link, Cartridge) (see 5.1).

Categories	Defects	AQL	Method of inspection
Major			
101	Improper level of packaging and packing (see procurement documents)	1.0	Visual
102	Incorrect or illegible marking	1.0	Visual
103	Inadequate cleaning and drying	1.0	Visual
104	Improper preservative application and drainage	1.0	Visual
105	Container interior contaminated with dirt, lint or other foreign matter	1.5	Visual
106	Incorrect quantity of links in container	1.5	Visual/Manual
107	Improper closure of shipping containers	1.5	Visual
Minor			
201	Workmanship	4.0	Visual

4.5.3.5 Additional examination. In addition to the examination specified in 4.5.3.1, a sample of 300 links shall be taken from each inspection lot regardless of lot size and visually examined for burrs, sharp edges and sharp corners. Failure of any link to meet the requirements shall cause rejection of the represented lot. Any link having burrs, sharp edges or sharp corners which would remove metal from the cartridge case shall be considered unacceptable.

4.5.4 Testing. In addition to the tests required by applicable drawings and specifications, the following tests shall apply. Unless otherwise specified herein, tests shall be performed on samples taken from each inspection lot. Test fixtures and equipment shall be in accordance with Government drawings or, in the absence of drawings, shall be subject to approval by the procuring agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

4.5.4.1 Material testing. Tests required by applicable drawings and specifications shall be performed on material used in the manufacture of links. The contractor shall furnish the Government representative with certified copies of chemical and check analysis and reports of physical tests as may be required for each lot of each material used. The contractor shall also furnish the Government representative with certification that the materials used in packaging conform to the applicable specifications.

4.5.4.2 Functioning testing. A sample of 300 links shall be taken from the quality evaluation sample (see 3.1) and a sample of 300 links from each month's production thereafter shall be tested for functioning by the testing agency specified in the contract using the test methods specified in 4.6.8. Links failing to function properly shall be inspected for compliance with requirements. Sample links will not be returned to the contractor except as evidence to support the testing agency's report.

4.5.4.3 Welded joint pull testing. Sampling shall be in accordance with MIL-STD-414 to determine compliance with the welded joint pull requirements of drawing C11577416 using the test method specified in 4.6.1. The sampling plan shall conform to Section B, Part I and Inspection Level III at an AQL of 0.065.

4.5.4.4 Elongation testing. A sample of 72 links (six belts of 12 sample links each) taken from each shift's production shall be tested for pull and permanent set using the test method specified in 4.6.2. If all belts meet the requirements, the represented shift's production shall be accepted. If two or more belts fail to meet the requirements, the represented shift's production shall be rejected. If one belt fails to meet the requirement, a retest shall be conducted using 144 additional links (12 belts of 12 sample links each). If the number of defective belts from both tests combined is two or more, the shift's production shall be rejected.

4.5.4.5 Hardness testing. Samples of links from each heat-treat lot, subject to the requirements of MIL-STD-105, using size code letter "G" regardless of the quantity in the heat-treat lot, shall be tested by the contractor for the hardness requirements using the test method specified in 4.6.3. Acceptance shall be based on an AQL of 0.40 percent.

4.5.4.6 Material soundness testing. Samples of links from each heat-treat lot, subject to the requirements of MIL-STD-105, using size code letter "G" regardless of the quantity in the heat-treat lot, shall be tested by the contractor for the material soundness requirements using the test method specified in 4.6.4. Acceptance shall be based on an AQL of 0.40 percent.

4.5.4.7 Protective finish testing. The accelerated corrosion test of links shall be made in accordance with drawing A7309999.

4.5.4.8 Cleanliness testing. Links shall be tested by the contractor for cleanliness using the test method specified in accordance with MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

4.5.4.9 Performance testing. Eight belts of 12 links each taken at random from each shift's production shall be tested by the contractor using test methods specified in 4.6.5, 4.6.6 and 4.6.7. If one or more links fail to meet the requirements, the represented shift's production shall be rejected.

4.5.5 Inspection equipment. Unless otherwise specified in procurement documents (see 6.1), responsibilities for acquisition, maintenance and disposition of measuring and testing equipment prescribed on lists contained on the List of Inspection Equipment, IEL 11577416, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607 and MIL-C-45662.

4.6 Test methods.

4.6.1 Welded joint test. Each cartridge link of the sample shall be pulled by means of two close-fitting pins, one inserted into the double loop and the other inserted into the single loop, until failure of the welded joint begins. The direction of the load shall be in accordance with the test requirement note of drawing C11577416. Failure is defined as a change in the welded joint as determined by visual examination. The failure load of each link in the sample shall be recorded. If a link fails to meet the requirement of the drawing by failure in a portion other than the welded joint, another link shall be substituted and tested.

4.6.1.1 When the failure loads of each link in the sample have been obtained, calculations shall be made in accordance with MIL-STD-414 to determine acceptability of the lot; however, any link in which the weld failed at less than the lower limit as specified in the test requirement note of drawing C11577416 shall cause automatic rejection of the lot. All test data and calculations shall accompany each lot submitted to the Government. The links used for this test shall be scrapped.

4.6.2 Elongation test. Belts of twelve sample links each shall be assembled with dummy cartridges using a tool to insure proper positioning of the cartridges in the links. These sample belts shall be tested for pull and permanent set (see 3.2.3) using Government approved inspection equipment. The belts shall be placed on the mandrels of the fixture, the specified initial load applied and then the indicator gage set to zero. The specified final load shall be applied at a uniform rate in a time interval of not less than two seconds. After the final load has been applied for the specified time, the load shall be reduced to the initial value and the indicator gage shall indicate whether the permanent set requirement has been met. Then the initial load shall be reduced to zero, the belt removed and visually examined for cracks, tears and fractures.

4.6.3 Hardness test. The sample links shall be tested for hardness requirements of drawing C11577416 using test method ASTM E18. All hardness testing shall be conducted prior to phosphating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

4.6.4 Material soundness test. The sample links shall be tested for material soundness requirements of drawing C11577416, using magnetic-particle inspection method and criteria per drawing B8769067. All magnetic-particle testing shall be conducted prior to phosphate coating.

4.6.5 Twisting test. Each sample belt, held vertically, shall be twisted manually at the lowest link in both clockwise and counterclockwise directions until the lowest cartridge forms an angle of 180 degrees with the uppermost cartridge. Failure of any belt to be twisted manually without force or failure of angles between individual cartridges being approximately equal shall be cause for rejection.

4.6.6 Fanning test. Each sample belt, placed on a flat surface shall be fanned manually in one direction so that a semicircle is formed by the noses of the cartridges, and shall be fanned in the opposite direction so that a semicircle is formed by the bases of the cartridges. Failure to accomplish either of the above manually without force shall be cause for rejection. In addition, failure to meet the dimensional requirements of the semicircles, as specified in 3.2.2, shall be cause for rejection.

4.6.7 Hinging and foldover test. Each cartridge link of each sample belt shall be tested as follows: Manually rotate the single loop of each link in the double loop of the link holding the cartridge. Failure of the single loop to rotate freely without effort on the cartridge shall be cause for rejection. Also each sample belt shall be tested as follows: Lay each belt on a flat surface to its full length, with the open side up, and draw one end back upon the remainder of the belt until the belt is completely reversed (open side of links down). Then, draw back one end of the reversed belt upon the remainder of the belt until the belt is returned to the original (open side of links up) position.

4.6.8 Functioning test. The sample links shall be assembled three times into belts using the 20MM TP-T Cartridge M206, and fired in bursts of 50 rounds in M139 automatic guns.

5. PREPARATION FOR DELIVERY

5.1 Links. Links shall be prepared for delivery in accordance with the applicable Packaging Data Sheet PDS 11577416 for the levels of packaging and packing specified in the procurement document.

5.2 Marking. Marking of unit packages and shipping containers shall be in accordance with MIL-STD-129.

5.3 Data cards. Ammunition data cards shall be furnished. The requirements of MIL-STD-1167 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Title, number and date of the Technical Data Package which contains the list of drawings, specifications, engineering parts lists and inspection equipment lists pertinent to the link, showing applicable revision dates.

6.1.2 Selection of applicable levels of preservation, packaging and packing.

6.1.3 Place of final inspection and acceptance.

6.1.4 Report of results of examination and tests shall be forwarded to the technical agency as prescribed in the contract.

6.1.5 When a quality evaluation sample is required the number of links, the testing agency and the requirements for submission by the contractor shall be specified.

6.1.6 Responsibility for acquisition, maintenance and disposition of acceptance inspection equipment, if other than that specified.

6.1.7 Responsibility for furnishing ammunition.

6.1.8 The agency responsible for gaging and function testing.

6.2 When action by a testing agency is required, work programming will be effected with the testing agency at the earliest practicable date.

6.3 When links are prepared for domestic shipment from manufacturers to ammunition loading plants, level C preservation, packaging and packing as specified on the applicable packaging data sheet shall be used.

6.4 When warranted, the following paragraph should be included in the written contract to cover the type of quality assurance system that is desirable for this item:

> Contractor's quality assurance system. The contractor shall provide and maintain a quality assurance system in accordance with MIL-I-45208.

6.5 Supersession data. This specification includes the requirements of Watervliet Arsenal Purchase Description WVTPD 1039A dated 29 March 1967.

Custodian:
Army—WC

Preparing activity:
Army—WC (Wvt Ars)

Project No. 1305-A579

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-L-45594(WC) Link, Cartridge, 20MM: XM24

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

To detach this form, cut along this line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
HEADQUARTERS
U. S. ARMY WEAPONS COMMAND

ROCK ISLAND, ILLINOIS 61201

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding General
U. S. Army Weapons Command
ATTN: AMSWE-RER-S
Rock Island, Illinois 61201

FOLD

To detach this form, cut along this line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:30Z
Check the source to verify that this is the current version before use.